

A Dual-Regime Faithfulness Audit of Explainers for House-Price Valuation Models

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Abstract—Deployed automated valuation models are gradient-boosted ensembles whose opacity forces practitioners to rely on a single post-hoc explainer, usually chosen by convenience and trusted without verification. We give a validated recipe for choosing one. The metrics normally used to verify an explanation, namely deletion, insertion, comprehensiveness and infidelity, replace features with a marginal background, which breaks feature dependencies and moves the input off the data manifold, so an audit can measure out-of-distribution behaviour rather than faithfulness. Our central contribution is a dual-regime faithfulness audit that scores five explainers under this marginal regime and under an on-manifold regime built from conditionally matched donor values, on three deployed ensembles spanning King County, California and Ames. The audit yields a criterion that is both stable and useful. Infidelity is regime-stable where the deletion ranking is not, whose cross-regime correlation is statistically indistinguishable from zero under a bootstrap over instances. Infidelity also predicts downstream actionability: acting on the top contributions of the explainers it endorses moves the predicted price in the attributed direction 71 to 77% of the time, against 47 to 53% for the common TreeSHAP default, which the audit independently flags as least faithful in every market. We expose the resulting faithfulness-weighted consensus, which is faithful, actionable and regime-stable, through a deployed service, and offer the audit as a reusable tool for auditing explanation tools.

Index Terms—explainable AI, faithfulness, out-of-distribution perturbation, Shapley values, ensemble learning, automated valuation, trustworthy machine learning

I. INTRODUCTION

Automated valuation models now support lending, taxation and risk assessment, and for tabular housing data the dominant tools are gradient-boosted tree ensembles, which capture the nonlinear and interaction-rich structure that linear hedonic models cannot represent [1]. Because valuation is a regulated and consequential task, these models are seldom deployed without an explanation layer, and in practice a single post-hoc explainer such as SHAP or LIME is attached and its attributions are read as the reasons behind a prediction [29]. Two assumptions in this routine deserve scrutiny. The first is that the chosen explainer is faithful to the model that is actually deployed. The second, less examined, is that the metrics used to check faithfulness are themselves valid.

A remaining challenge is that the standard faithfulness metrics, including deletion and insertion curves and infidelity, perturb an input by replacing selected features with a marginal background, usually a population mean or an unconditional sample. This procedure breaks the dependence structure among features and moves the perturbed point off the data manifold, into a region the model rarely or never observed during training. The recorded change in the model output then reflects out-of-distribution behaviour as much as the importance of the removed features [23]. The faithfulness score, and the ranking of explainers it induces, may consequently be an artefact of the perturbation design rather than a property of the explanations. This concern is acute for deployed valuation models, where the deployed function is typically a blend of several learners over a transformed target, so a convenient tree-native explainer can quietly explain a component rather than the whole.

To address this limitation we conduct a dual-regime faithfulness audit. We fix a deployed ensemble and score five explanation methods under two perturbation regimes: a standard marginal regime, and an on-manifold regime that replaces masked features with values drawn from a conditionally matched background donor, keeping the perturbed input within the data distribution. We run this audit on three markets, a primary King County study with California and Ames replications, in order to separate findings that generalise from those specific to one city. The central questions are therefore three. The first concerns reliability of measurement: whether the ranking of explainers by faithfulness is stable when the perturbation moves from marginal to on-manifold, or whether the apparent best method is regime-dependent. The second concerns the default: whether the tree-native explainer remains faithful to a blended, trend-adjusted deployed model, as measured by infidelity, under both regimes and in all three markets. The third is constructive: whether faithfulness, once measured, can be used to combine explainers into a consensus that is robust to the choice of regime and provides more interpretable and actionable explanations.

The contributions are as follows. Our central contribution is a dual-regime audit that exposes the dependence of rank-based faithfulness metrics on the perturbation manifold: with

consistent evidence across three markets, and bootstrap confidence intervals, we show that these metrics do not preserve the ranking of explainers across regimes, whereas infidelity does. Building on this, we provide evidence that the TreeSHAP default is the least faithful method as measured by infidelity in both regimes and all three markets, even after a blend-aware correction. We therefore propose a faithfulness-weighted consensus that remains among the most faithful explanations under both regimes, and we deploy it in a service that returns valuations together with interpretable and actionable explanations. Figure 1 summarises the design.

II. RELATED WORK

Hedonic price theory [1] and spatial econometrics [2] give interpretable but linear baselines for valuation. In practice they are outperformed by gradient boosting, so modern automated valuation is built on XGBoost [3], LightGBM [4] and CatBoost [5], frequently combined into a single ensemble [7] as in recent house-price systems [6]. Because the resulting model is opaque, it is increasingly paired with a post-hoc explainer to recover the determinants of price [30]. The explainer in this pairing, however, is usually chosen with far less care than the model it explains, and it is this asymmetry that our work addresses.

The explainers we audit are individually well founded, yet they rest on different principles and so need not agree. SHAP [8] grounds an attribution in the Shapley value [9]: KernelSHAP estimates that value model-agnostically, while Shapley sampling [10] provides an independent estimator. TreeSHAP [11] instead computes exact tree attributions, which is fast but ties the explanation to the model’s internal structure. LIME [12] fits a local linear surrogate, and occlusion [13] measures the effect of removing one feature at a time. Since these methods optimise different objectives, their disagreement is structural rather than incidental, and is documented as a practical problem [19]. The ambiguity is sharpest for the Shapley family, whose value depends on the chosen reference distribution, so that one model admits many different attributions [21] with known shortcomings as importance scores [20]. Which explanation to trust is therefore an open question [22], and we treat it as an empirical one.

How to evaluate an explanation is itself unsettled [35], and even the meaning of faithfulness is debated, with calls for graded rather than binary criteria [36]. The available metrics fall into two groups. The first scores an attribution by perturbing the input in order of attributed importance, as in deletion and insertion curves [14] and the comprehensiveness and sufficiency measures of ERASER [15]. The second measures additive fidelity directly, through infidelity and maximum sensitivity [16], alongside analyses of how stable the methods themselves are [17]; aggregating several explanations has also been proposed as a way to reduce this variance [18]. Sanity checks have even shown some attribution methods to be insensitive to the model they purport to explain [34], which is exactly the failure a faithfulness audit must rule out. The motivation for our work is that

almost all of these metrics share one hidden assumption: to remove a feature they replace it with a marginal background, as if features were independent. Aas et al. [23] showed that this assumption yields misleading Shapley estimates when features are correlated and proposed conditional, on-manifold alternatives; the idea was later developed into explanations that respect the data manifold [25] and framed as a choice between being true to the model and true to the data [27]. The same off-manifold concern surfaces in removal-based explanation [26], in remove-and-retrain analyses [24], and most pointedly where off-manifold queries let an adversary hide a biased model from its explainer [28]. These concerns, though, have been raised almost entirely about the explainers. Our contribution is to carry the same scrutiny into the evaluation itself: we audit five explainers under both a marginal and an on-manifold regime on three deployed valuation models, and then use the more regime-stable criterion to combine them.

III. DEPLOYED ENSEMBLES IN THREE MARKETS

Our primary study uses the public King County, Washington home-sales record [31], comprising 553,655 cleaned transactions over 318,981 unique properties from 1999 to 2025, described by 57 engineered features. Identifiers are excluded from the model matrix to prevent memorisation, and the target is log-price modelled as a residual over an extrapolated median market trend. The evaluation protocol is strict (Table I): every property in the 2024 to 2025 benchmark is purged from training, which leaves exactly zero train and test property overlap and ensures that accuracy and, more importantly, explanations are not artefacts of repeat sales. The deployed model is a convex blend of LightGBM, XGBoost and CatBoost with weights learned on a property-disjoint 2023 validation set by sequential least-squares programming,

$$\min_{w \geq 0, \mathbf{1}^\top w = 1} \sqrt{\frac{1}{n} \sum_i (y_i - \sum_k w_k \hat{y}_i^{(k)})^2}, \quad (1)$$

giving weights 0.666, 0.223 and 0.111. Table II reports accuracy on the future benchmark, where the blend reaches $R^2 = 0.826$ and $\text{RMSLE} = 0.200$. We report this only to establish a credible substrate whose explanations are worth auditing, not as a contribution in itself.

We use three markets deliberately, choosing markets that differ in scale, region, feature structure and even unit of analysis, so that a finding which recurs across all three is unlikely to be specific to one dataset. King County is the large, transaction-level primary study that supports a strict leakage-controlled split; California and Ames are widely cited public benchmarks that stress the audit in different directions. We therefore treat the three together as a generalisation test rather than a like-for-like replication, and revisit the resulting differences in split protocol and granularity as limitations.

The second market addresses the external-validity limitation of a single-market study: we replicate the design on the California housing record of Pace and Barry [33], a widely cited benchmark of 20,640 block-group records, aggregated rather than individual sales, with eight base attributes, to

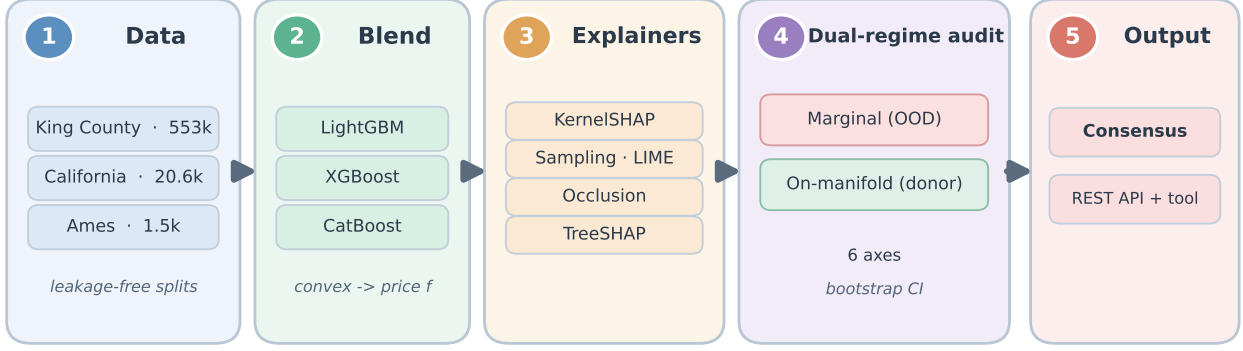


Fig. 1: Overview of the study. A leakage-controlled boosting blend is deployed in each market; five explanation methods are scored on the deployed function under two perturbation regimes, marginal and on-manifold; the audit informs an infidelity-grounded consensus that is served through an API and a chatbot.

TABLE I: King County temporal property-disjoint split, with zero train and test overlap.

Split	Rows	Years	Purpose
Tuning	464,541	≤ 2022	base hyper-tuning
Validation	12,797	2023	blend weights
Final train	490,797	≤ 2023	fit deployed model
Benchmark	30,099	2024 to 2025	evaluation

which we add the standard rooms-per-household, bedrooms-per-room and population-per-household ratios. The same three learners are blended by Eq. (1); here CatBoost dominates the blend, taking weight 0.897 while XGBoost and LightGBM take 0.085 and 0.018, and the blend attains $R^2 = 0.723$ on a held-out split. This market is deliberately different in scale, granularity and feature structure, which makes it a useful test of whether the audit findings generalise. The contrast in blend composition also matters for the audit, since TreeSHAP will be applied to a LightGBM component that carries almost no weight in the deployed California model.

The third market is the Ames, Iowa housing record of De Cock [32], a widely used benchmark of 1,460 residential sales with rich structural detail. To keep the audit comparable across markets we model the twenty most informative numeric attributes, selected by a preliminary importance ranking, and blend the three learners as before. The deployed blend again places most weight on CatBoost (0.897) with the remainder on LightGBM, and attains $R^2 = 0.895$ on a held-out split. Ames differs from the two coastal markets in both region and housing stock, so a finding that recurs here is unlikely to be an artefact of a single market.

A subgroup analysis on the zero-overlap King County benchmark (Table III) behaves as a credible design should: genuinely new properties, appearing for the first time in the benchmark, are harder to value (RMSLE 0.232) than previously observed ones (0.180), and accuracy drifts mildly

TABLE II: Deployed-model accuracy. King County is evaluated on the strict 2024 to 2025 benchmark; California and Ames on held-out splits. The blend is reported as the deployed substrate, not as a contribution.

Market	Model	RMSLE	RMSE	R^2
King County	Convex blend	0.2004	308,774	0.8261
	XGBoost	0.2022	312,324	0.8221
	LightGBM	0.2020	312,681	0.8217
	Multiple linear	0.4654	653,532	0.2210
California	Convex blend	0.2841	63,352	0.7230
	best single learner	0.2842	64,600	0.712
Ames (Iowa)	Convex blend	0.1303	28,014	0.8950
	best single learner	0.1308	28,600	0.889

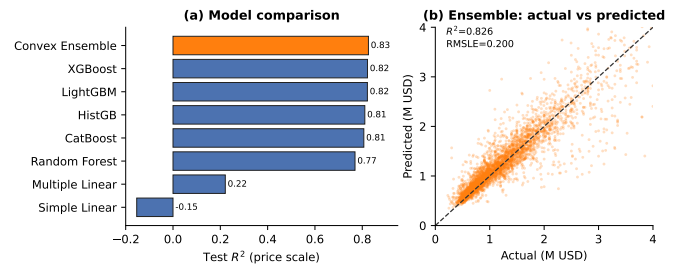


Fig. 2: King County deployed substrate. (a) Test R^2 across candidate models, where the convex blend leads and the linear baselines collapse under temporal extrapolation. (b) Actual versus predicted price for the deployed blend.

from 2024 to 2025. Because no benchmark property occurs in training, none of this reflects memorisation, which is the precondition for reading the explanations as statements about the model rather than about leaked history.

IV. DUAL-REGIME FAITHFULNESS AUDIT

For a deployed function f , let $\mu = \mathbb{E}_{b \sim \mathcal{B}}[f(b)]$ be the mean prediction over a fixed background $\mathcal{B}$ and $g(x) = f(x) - \mu$

TABLE III: King County subgroup audit on the zero-overlap benchmark.

Subgroup	Rows	RMSLE	RMSE	R^2
Year 2024	15,133	0.1875	290,257	0.8439
Year 2025	14,966	0.2127	326,431	0.8087
Previously observed	19,045	0.1799	282,448	0.8509
First appearance	11,054	0.2315	349,510	0.7859

the quantity that additive attributions explain. Each method produces an attribution vector $\phi(x)$ for the same f , the same background and the same price-stratified instances, so that any difference reflects the method and not the setup. The agnostic methods are KernelSHAP and Shapley sampling [8], LIME [12] and occlusion [13]; the fifth, TreeSHAP [11], explains the dominant or, in California, a low-weight LightGBM component, and represents the fast default that practitioners use.

We score each method on six axes. Ranking features by $|\phi|$, deletion area averages $g(x_t)/g(x)$ as top features are masked, insertion area reverses this, and comprehensiveness [15] is the share of $g(x)$ removed when the top fifth of features are ablated. Infidelity [16] is $\mathbb{E}_I[(I^\top \phi - (f(x) - f(x_{-I})))^2]$ under random subset perturbations, with each ϕ additively calibrated to $g(x)$. Stability [16] is the relative change of ϕ under small input noise, and cost is time per instance. The contribution of this paper is to evaluate these axes under two perturbation regimes. In the *marginal* regime a masked feature is set to the background mean, the standard but off-manifold choice. In the *on-manifold* regime the masked features are set to the values of a background donor whose retained features are closest to the instance, so that the perturbed point remains a plausible, in-distribution property, following the conditional logic of Aas et al. [23]. Concretely, given an instance x and a set K of retained features, the donor is the background row b^* that minimises the standardised distance $\sum_{j \in K} (x_j - b_j)^2 / \sigma_j^2$, where σ_j is the pooled deviation of feature j , and the masked features are then set to b^* on the complement of K . As features are progressively masked along an attribution ranking, the conditioning set K shrinks and the donor is reselected at each step, so the perturbed point tracks the manifold rather than collapsing toward a global mean. We verify the construction is less out of distribution: under 50% masking on King County, the donor perturbations are closer to their nearest background neighbour than marginal mean-replacement for every audited instance (3.7 against 5.1 in standardised units) (Fig. 5), so the regime queries the model in a region it is more likely to have seen. The two regimes differ only in this replacement rule; the attributions, the background and the instances are held fixed, which isolates the effect of the perturbation on the measured faithfulness.

A remaining question is how to combine the methods. We weight each agnostic method by inverse infidelity, the axis that measures additive fidelity directly and, as the results show, is

Algorithm 1 Dual-regime faithfulness audit

Require: deployed f , background $\mathcal{B}$, instances X , explainers $\{\phi_m\}$, perturbations T

- 1: $\mu \leftarrow \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} f(b)$; $\Phi_m \leftarrow \phi_m(X)$ for all m ▷ attributions once
- 2: **for** regime $r \in \{\text{MARG}, \text{ONMAN}\}$, method m **do**
- 3: $\pi \leftarrow \text{ARGSORT}(|\Phi_m|, \text{desc})$
- 4: **for** $k = 0$ to p **do** ▷ deletion curve
- 5: $X' \leftarrow \text{MASK}(X, \pi_{1:k}, r)$; $c_k \leftarrow (f(X') - \mu) / (f(X) - \mu)$
- 6: $\text{Del}_m^r \leftarrow \text{mean}_k c_k$
- 7: **for** $t = 1$ to T **do** ▷ infidelity
- 8: sample subset I ; $X' \leftarrow \text{MASK}(X, I, r)$
- 9: $e_t \leftarrow (\Phi_m^\top \mathbb{1}_I - (f(X) - f(X'))^2) / (f(X) - \mu)^2$
- 10: $\text{Infid}_m^r \leftarrow \text{mean}_t e_t$
- 11: **function** $\text{MASK}(X, S, r)$
- 12: **if** $r = \text{MARG}$ **then return** X with features $S \rightarrow$ background mean
- 13: **else return** X with $S \rightarrow$ donor b^* , $b^* = \arg \min_{b \in \mathcal{B}} \sum_{j \notin S} (x_j - b_j)^2 / \sigma_j^2$

the more stable criterion across regimes,

$$w_m = \frac{1/\text{INFD}_m}{\sum_{m' \in \mathcal{A}} 1/\text{INFD}_{m'}}, \quad \phi^{\text{FWC}}(x) = \sum_{m \in \mathcal{A}} w_m \tilde{\phi}_m(x), \quad (2)$$

with each $\tilde{\phi}_m$ calibrated to $g(x)$, so the convex combination is itself additive. TreeSHAP is excluded from the fusion as it targets a different quantity. We therefore propose this faithfulness-weighted consensus as an explainer that does not require committing in advance to a single method.

For each market the audit fixes a background of 40 transactions and a set of price-stratified instances, 60 for King County and 40 for California, so that the audit spans the price range. KernelSHAP, Shapley sampling and LIME use a few hundred to a thousand samples per instance, while occlusion and TreeSHAP are exact given the background. Infidelity is estimated from random subset perturbations at a mask probability of 0.3, and stability from small Gaussian perturbations of the continuous features. In the on-manifold regime, the donor for a masked subset is the background row whose retained features, standardised by their pooled deviation, are closest in Euclidean distance to the instance. All attributions are computed once and reused across both regimes so that the comparison is controlled. Because each axis ranks only a handful of explainers, a single cross-regime Spearman correlation would be coarse, so rather than a point estimate we resample the audited instances with replacement (2,000 draws per market) and report the interval of the correlation. The uncertainty is thus over the instances that drive the ranking, not over the few methods, which keeps the comparison honest despite the small number of explainers.

V. AN EXPLANATION TOOL

The audit is useful only if it changes practice, so we expose the deployed ensemble and the consensus through a small

TABLE IV: King County cross-method agreement. Upper triangle gives Spearman rank correlation of global importance, lower triangle (italic) gives top-ten Jaccard overlap.

	KS	SS	LI	OC	TR
KS	.	0.97	0.49	0.95	0.80
SS	0.82	.	0.46	0.97	0.83
LI	0.67	0.54	.	0.46	0.66
OC	0.67	0.67	0.54	.	0.80
TR	0.67	0.54	0.82	0.43	.

service. A REST API offers `/predict` for a valuation, `/explain` for the additive consensus attribution in USD, and `/chat` for a plain-language rationale, optionally through a large language model and otherwise through a deterministic offline summary. Because the consensus is additive and calibrated to the model output, the rationale is anchored in the audited explanation rather than in a single unverified method, which is what we mean by an explanation that is both faithful and actionable. Figure 3 sets out the request workflow, with each component listing its inputs and outputs.

Figure 4 shows the service in operation on a representative property, returning the price, the audited consensus contributions in USD and a short rationale.

The service is self-contained: without an API key the prediction and the consensus explanation remain fully functional through an offline summary, and with a key the assistant adds a natural-language rationale. All datasets are public, the splits are fixed, and every reported number is produced by executing the same pipeline, so the experiments reproduce end to end.

VI. RESULTS

The explainers disagree on the drivers of price. On King County, the agnostic Shapley methods and occlusion concentrate importance on living area, whereas LIME ranks transaction year first. As measured by Spearman rank correlation of global importance, KernelSHAP, Shapley sampling and occlusion agree at 0.95 to 0.97, while LIME agrees with them at only 0.46 to 0.49, with a top-ten feature overlap of 0.54 by Jaccard. The account given of a valuation therefore depends on the explainer, even though all methods explain the same deployed model.

Rank-based faithfulness is unstable across regimes. Under the marginal regime alone, no single method is best on every axis (Table V): LIME and occlusion lead the rank-based curves but trail on infidelity, while the Shapley family is the most additively faithful. This already cautions against selecting an explainer by a single axis. The dual-regime audit makes the concern precise. Table VI reports deletion and infidelity under both regimes, and Following the bootstrap protocol above, we report intervals of the cross-regime correlation rather than a point value. In all three markets the deletion correlation is statistically indistinguishable from zero, with confidence intervals that span it (King County $[-0.68, 0.61]$, California $[-0.89, 0.60]$, Ames $[-0.37, 0.83]$), so its sign is not even reliable and the deletion ranking carries essentially

no regime-invariant information. The infidelity correlation is consistently higher and bounded away from or near the top of the range (King County $[0.82, 1.00]$, California $[0.66, 0.94]$, Ames $[0.37, 1.00]$). The robust signal is the paired contrast: the infidelity ranking is more stable across regimes than the deletion ranking in 100%, 99% and 94% of resamples for the three markets. The reshuffling behind the deletion result is concrete; on King County, occlusion moves from a deletion of 0.041, among the best, to 0.287, the worst, when the perturbation is placed on the manifold. Selecting an explainer by deletion curves is therefore unreliable, while infidelity provides a regime-stable criterion.

The default explainer is unfaithful even after a fair correction. As measured by infidelity, TreeSHAP is the least faithful method in every column of Table VI. A natural objection is that the comparison is unfair, because the default TreeSHAP explains the LightGBM component rather than the deployed blend. We therefore construct a blend-aware TreeSHAP that sums the exact tree attributions of the three base learners weighted by the blend coefficients, so that it targets the same function the agnostic methods explain. This fair variant does reduce infidelity, from about 9.7 to about 6.6 on King County, which confirms that part of the original gap was attributable to the mismatch. The reduced value nonetheless remains several times the infidelity of the agnostic estimators, which sit near 1.0 to 2.0. The residual gap is structural: exact tree attributions decompose the model along the tree partition rather than along the additive price response that infidelity measures, and the conclusion is invariant to the perturbation regime. We read this as evidence that a tree-native attribution should not be assumed faithful to a blended deployed model without verification, even when it is made blend-aware.

The results generalise to California and Ames. The bootstrap intervals above already establish that the deletion instability and the comparative stability of infidelity recur in both replication markets. The TreeSHAP result is sharper still: in both, the deployed blend places weight 0.897 on CatBoost and little on the LightGBM component that TreeSHAP explains, so the explainer describes a part of the model that contributes almost nothing to the deployed prediction, and its infidelity is the highest of all methods under both regimes, reaching 12.66 and 34.55 on California and 33.16 and 41.51 on Ames. We read the three-market evidence as consistent rather than identical: the precise reshuffling of rank-based axes is market-dependent, while the instability of those axes, the comparative stability of infidelity, and the unfaithfulness of the tree-native default are common to all, which suggests these are properties of the audit and of the deployed-model structure rather than of a single dataset.

The faithfulness-weighted consensus is regime-robust. With King County weights 0.32, 0.32, 0.21 and 0.15, it is constructed from infidelity, so to avoid a circular evaluation we judge it on the independent axes. On the rank-based axes it attains the best deletion and comprehensiveness on King County (Table V), and on infidelity it remains among the most faithful under both regimes in all three markets,

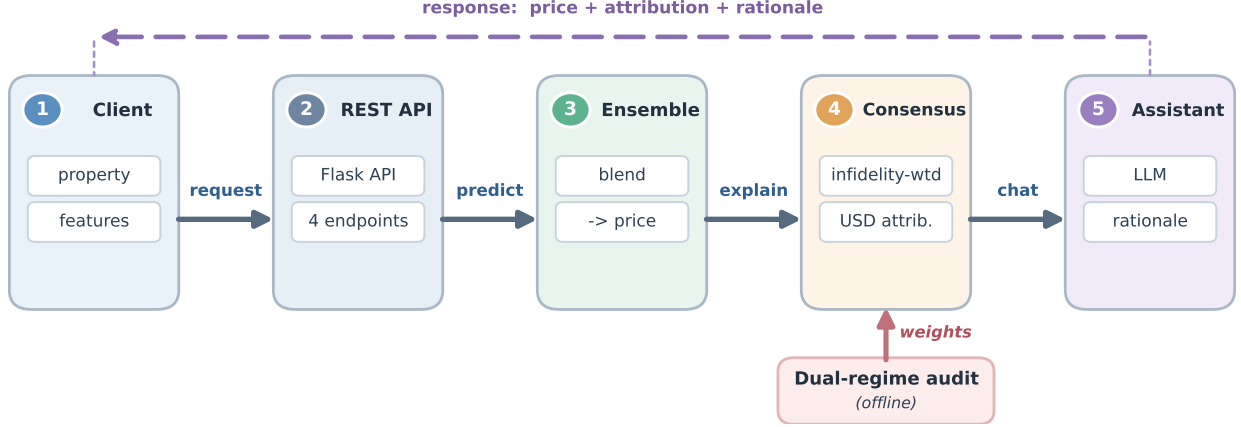


Fig. 3: Request workflow of the explanation service. A client request flows through the Flask API to the ensemble (predict), the consensus explainer (explain) and the assistant (chat); the response returns the price, the additive attribution and the rationale. Each block lists its inputs and outputs, and the consensus weights are set offline by the dual-regime audit.

TABLE V: Marginal-regime audit on King County, reported per axis. Arrows give the preferred direction; **bold** and underline mark the best and second best per axis. We deliberately avoid a single aggregate score across axes of differing nature.

Method	Deletion ↓	Insertion ↑	Comprehens. ↑	Infidelity ↓	Stability ↓	Cost (s) ↓
KernelSHAP	0.085	0.924	0.877	0.929	0.283	1.648
Shapley sampling	0.055	0.943	0.974	0.922	0.430	1.056
LIME	<u>0.031</u>	1.026	1.001	1.448	0.372	0.336
Occlusion	0.041	0.976	<u>1.038</u>	1.931	0.376	0.025
TreeSHAP	0.090	0.906	<u>0.886</u>	9.066	<u>0.284</u>	0.177
Consensus (FWC, ours)	0.001	<u>0.989</u>	1.090	<u>1.022</u>	0.364	3.065

TABLE VI: Dual-regime audit, reported in detail for King County and California; Ames is summarised in the text and Fig. 8. Deletion and infidelity are reported under the marginal (M) and on-manifold (OM) regimes. The bottom row gives the Spearman rank correlation between the two regimes: deletion rankings do not survive the change of regime, whereas infidelity is more stable. TreeSHAP is the least faithful method by infidelity in every column.

Method	King County				California			
	Del. M	Del. OM	Inf. M	Inf. OM	Del. M	Del. OM	Inf. M	Inf. OM
KernelSHAP	0.085	0.112	0.937	1.137	0.333	0.764	8.78	13.99
Shapley sampling	0.055	0.014	0.904	1.249	0.395	0.838	7.33	29.46
LIME	0.031	0.006	1.461	1.579	0.497	0.894	9.65	29.74
Occlusion	0.041	0.287	1.906	2.265	0.400	0.202	10.34	28.20
TreeSHAP	0.090	0.012	9.066	8.383	0.373	0.997	12.66	34.55
Consensus (ours)	0.001	0.077	0.987	1.323	0.408	0.644	4.72	15.20
rank corr. (M,OM)	−0.03		+0.94		−0.14		+0.60	

ranking first on Ames and first or second on California. Figure 8 summarises the generalisation: in all three markets the deletion ranking is unstable across regimes while infidelity is comparatively stable, which is the property the consensus is built to exploit. We claim, modestly, that it provides a more stable basis for explanation than any single method selected by rank-based curves.

A faithful explanation should also be actionable: intervening on the features it ranks highest should move the prediction in the predicted direction. Setting each of an instance’s top features to its on-manifold donor value and checking the sign of the resulting change in predicted price (Fig. 7), the audit’s

verdict transfers cleanly. The explainer it endorses by infidelity, KernelSHAP and Shapley sampling, are directionally correct for 71 to 77% of top contributions, and the consensus for 64 to 72%, all well above the TreeSHAP default at 47 to 53%, which the audit independently ranks least faithful. Infidelity, the criterion our audit privileges, thus predicts which explanations can be acted upon, while the convenient default cannot.

VII. DISCUSSION

The results carry two practical implications. First, the validity of a faithfulness audit depends on its perturbation: because

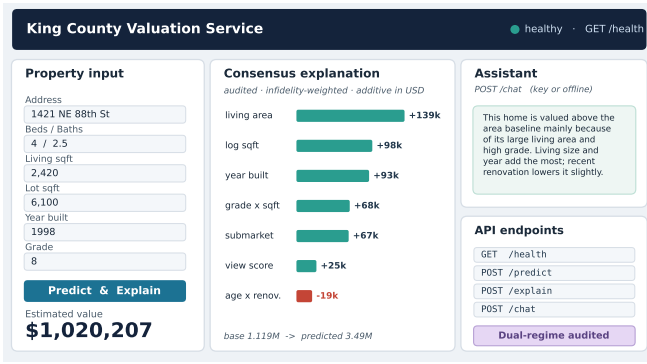


Fig. 4: The valuation service in operation. A single screen exposes the four endpoints: the input panel and predicted price, the audited and infidelity-weighted consensus contributions in USD, the plain-language rationale, and the service status.

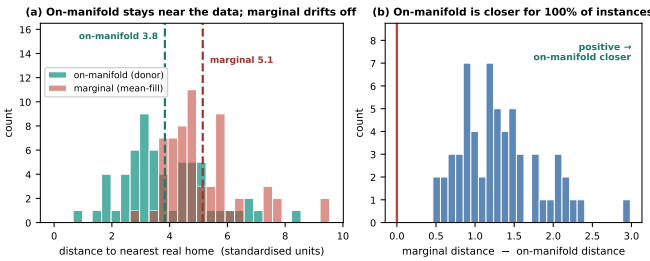


Fig. 5: Why the regime matters on King County. (a) Distance to the nearest real home after perturbing half the features: on-manifold donor replacement stays close to the data (3.8 standardised units), whereas marginal mean-replacement drifts away (5.1). (b) The donor is closer than the marginal background for every audited instance, so the marginal regime queries the model off the data manifold.

marginal perturbations move inputs off the data manifold, the rank-based curves they produce are regime-dependent, so an explainer selected by deletion may owe its apparent quality to the measurement rather than to its fidelity. We therefore recommend on-manifold auditing and, where a single criterion is needed, infidelity. Second, the convenient tree-native default should not be assumed faithful to a blended deployed model, since TreeSHAP is the least faithful method by infidelity in every setting we examined. The consensus weights agnostic explainers by their measured infidelity and so avoids committing to one method in advance.

More broadly, deletion, insertion and comprehensiveness are now treated as near-standard yardsticks, yet the ranking they induce can reverse under a more faithful perturbation; the practical lesson is to report the regime explicitly, prefer on-manifold constructions when feature dependence is strong, and anchor any single-number comparison on infidelity.

Several limitations qualify these claims. The study covers three markets and a finite set of instances. The on-manifold regime is one construction of a conditional perturbation among several, but the conclusions are not sensitive to it: replacing the single nearest donor with a k -nearest-donor average ($k = 3$ and 5) leaves the actionability ordering unchanged, with the

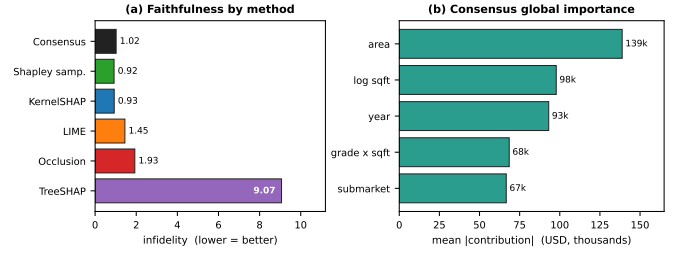


Fig. 6: King County. (a) Infidelity by method under the marginal regime, where TreeSHAP is the clear outlier and the consensus sits with the most faithful agnostic estimators. (b) Consensus global feature importance in USD, led by area, living size and year, consistent with hedonic intuition. The attributions are predictive associations, not causal effects.

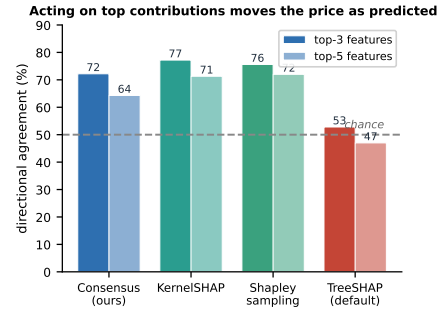


Fig. 7: Actionability on King County: the fraction of top-feature interventions (set to on-manifold donor values) whose effect on the predicted price agrees in sign with the attribution. The explainers the audit endorses move the price as predicted well above chance; the TreeSHAP default sits at chance.

audit’s endorsed methods staying well above the TreeSHAP default and the reference staying closer to real homes than the marginal one; richer generative samplers remain for future work. The evaluation protocol is also not uniform: King County uses a strict temporal, property-disjoint split, whereas the two replication markets use random splits, and the California target is capped at its upper bound, which affects both the model and its explanations. The infidelity values are not comparable in absolute terms across markets, since they depend on the price scale and feature count, so we rely on rankings and on within-market comparisons. The attributions are predictive associations and must not be read as causal effects, and the natural-language rationale produced by the service inherits the faithfulness of the consensus it summarises but is not itself audited. We mitigate these threats through leakage control, price-stratified sampling, two replication markets, and the reporting of every figure from executed code.

Finally, the audit is best understood as a reusable diagnostic rather than a result tied to housing. Its inputs are a deployed predictor, a background sample and a set of attributions, and its output is the per-axis, two-regime comparison of Table VI. Nothing in the procedure is specific to valuation, so it applies wherever a tabular model is deployed with a post-hoc explainer; the contribution is thus a tool for auditing

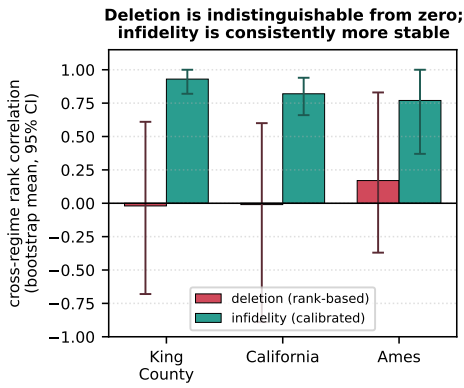


Fig. 8: Cross-market generalisation. Bootstrap mean and 95% confidence interval of the cross-regime rank correlation, by axis. In all three markets the deletion interval spans zero, so its ranking carries no regime-invariant information, whereas the infidelity interval stays high.

explanation tools. A natural extension is to broaden the on-manifold construction beyond a single conditional donor and to add further deployed markets.

VIII. CONCLUSION

We presented a dual-regime faithfulness audit of post-hoc explanations for deployed house-price valuation ensembles, on King County and replicated on California and Ames. Rank-based faithfulness metrics do not preserve the ranking of explainers when the perturbation moves from a marginal to an on-manifold regime, while infidelity is more stable, and the TreeSHAP default is the least faithful method by infidelity in both regimes and all three markets. We proposed a faithfulness-weighted consensus that remains among the most faithful explanations under both regimes and deployed it in a service. Extending on-manifold auditing to additional markets and to causal notions of explanation is left to future work.

REFERENCES

- [1] S. Rosen, “Hedonic prices and implicit markets: product differentiation in pure competition,” *J. Polit. Econ.*, vol. 82, no. 1, pp. 34-55, 1974.
- [2] L. Anselin, *Spatial Econometrics: Methods and Models*. Dordrecht: Kluwer, 1988.
- [3] T. Chen and C. Guestrin, “XGBoost: a scalable tree boosting system,” in *Proc. ACM SIGKDD*, 2016, pp. 785-794.
- [4] G. Ke et al., “LightGBM: a highly efficient gradient boosting decision tree,” in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 3146-3154.
- [5] L. Prokhorenkova et al., “CatBoost: unbiased boosting with categorical features,” in *Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 6638-6648.
- [6] R. Sibindi, R. W. Mwangi, and A. G. Waititu, “A boosting ensemble learning based hybrid light gradient boosting machine and extreme gradient boosting model for predicting house prices,” *Eng. Rep.*, vol. 5, no. 4, p. e12599, 2023.
- [7] D. H. Wolpert, “Stacked generalization,” *Neural Netw.*, vol. 5, no. 2, pp. 241-259, 1992.
- [8] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 4765-4774.
- [9] L. S. Shapley, “A value for n -person games,” in *Contributions to the Theory of Games II*. Princeton Univ. Press, 1953, pp. 307-317.

- [10] E. Štrumbelj and I. Kononenko, “Explaining prediction models and individual predictions with feature contributions,” *Knowl. Inf. Syst.*, vol. 41, no. 3, pp. 647-665, 2014.
- [11] S. M. Lundberg et al., “From local explanations to global understanding with explainable AI for trees,” *Nat. Mach. Intell.*, vol. 2, no. 1, pp. 56-67, 2020.
- [12] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why should I trust you? Explaining the predictions of any classifier,” in *Proc. ACM SIGKDD*, 2016, pp. 1135-1144.
- [13] M. Ancona, E. Ceolini, C. Öztireli, and M. Gross, “Towards better understanding of gradient-based attribution methods for deep networks,” in *Proc. Int. Conf. Learn. Represent.*, 2018.
- [14] V. Petsiuk, A. Das, and K. Saenko, “RISE: randomized input sampling for explanation of black-box models,” in *Proc. British Mach. Vis. Conf.*, 2018.
- [15] J. DeYoung et al., “ERASER: a benchmark to evaluate rationalized NLP models,” in *Proc. Assoc. Comput. Linguist.*, 2020, pp. 4443-4458.
- [16] C.-K. Yeh, C.-Y. Hsieh, A. S. Suggala, D. I. Inouye, and P. Ravikumar, “On the (in)fidelity and sensitivity of explanations,” in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 10965-10976.
- [17] D. Alvarez-Melis and T. S. Jaakkola, “On the robustness of interpretability methods,” in *Proc. ICML Workshop Human Interpretability Mach. Learn.*, 2018.
- [18] U. Bhatt, A. Weller, and J. M. F. Moura, “Evaluating and aggregating feature-based model explanations,” in *Proc. Int. Joint Conf. Artif. Intell.*, 2020, pp. 3016-3022.
- [19] S. Krishna et al., “The disagreement problem in explainable machine learning: a practitioner’s perspective,” *arXiv:2202.01602*, 2022.
- [20] I. E. Kumar, S. Venkatasubramanian, C. Scheidegger, and S. Friedler, “Problems with Shapley-value-based explanations as feature importance measures,” in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 5491-5500.
- [21] M. Sundararajan and A. Najmi, “The many Shapley values for model explanation,” in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 9269-9278.
- [22] T. Han, S. Srinivas, and H. Lakkaraju, “Which explanation should I choose? A function approximation perspective to characterizing post hoc explanations,” in *Adv. Neural Inf. Process. Syst.*, 2022, pp. 5256-5268.
- [23] K. Aas, M. Jullum, and A. Løland, “Explaining individual predictions when features are dependent: more accurate approximations to Shapley values,” *Artif. Intell.*, vol. 298, p. 103502, 2021.
- [24] S. Hooker, D. Erhan, P.-J. Kindermans, and B. Kim, “A benchmark for interpretability methods in deep neural networks,” in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 9737-9748.
- [25] C. Frye, D. de Mijolla, T. Begley, L. Cowton, M. Stanley, and I. Feige, “Shapley explainability on the data manifold,” in *Proc. Int. Conf. Learn. Represent.*, 2021.
- [26] I. Covert, S. Lundberg, and S.-I. Lee, “Explaining by removing: A unified framework for model explanation,” *J. Mach. Learn. Res.*, vol. 22, no. 209, pp. 1-90, 2021.
- [27] H. Chen, J. D. Janizek, S. Lundberg, and S.-I. Lee, “True to the model or true to the data?” *arXiv:2006.16234*, 2020.
- [28] D. Slack, S. Hilgard, E. Jia, S. Singh, and H. Lakkaraju, “Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods,” in *Proc. AAAI/ACM Conf. AI, Ethics, and Society*, 2020, pp. 180-186.
- [29] A. Barredo Arrieta et al., “Explainable Artificial Intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI,” *Inf. Fusion*, vol. 58, pp. 82-115, 2020.
- [30] S. Jin, H. Zheng, N. Marantz, and A. Roy, “Understanding the effects of socioeconomic factors on housing price appreciation using explainable AI,” *Appl. Geogr.*, vol. 169, p. 103339, 2024.
- [31] A. Krause, “King County (WA) home sales,” public dataset, accessed 2026.
- [32] D. De Cock, “Ames, Iowa: alternative to the Boston housing data as an end of semester regression project,” *J. Stat. Educ.*, vol. 19, no. 3, 2011.
- [33] R. K. Pace and R. Barry, “Sparse spatial autoregressions,” *Stat. Probab. Lett.*, vol. 33, no. 3, pp. 291-297, 1997.
- [34] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, “Sanity checks for saliency maps,” in *Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 9505-9515.
- [35] F. Doshi-Velez and B. Kim, “Towards a rigorous science of interpretable machine learning,” *arXiv:1702.08608*, 2017.
- [36] A. Jacovi and Y. Goldberg, “Towards faithfully interpretable NLP systems: how should we define and evaluate faithfulness?” in *Proc. Assoc. Comput. Linguist.*, 2020, pp. 4198-4205.